

AI in Healthcare: From Diagnosis to Decision

TRANSFORMING THE FUTURE OF PATIENT CARE

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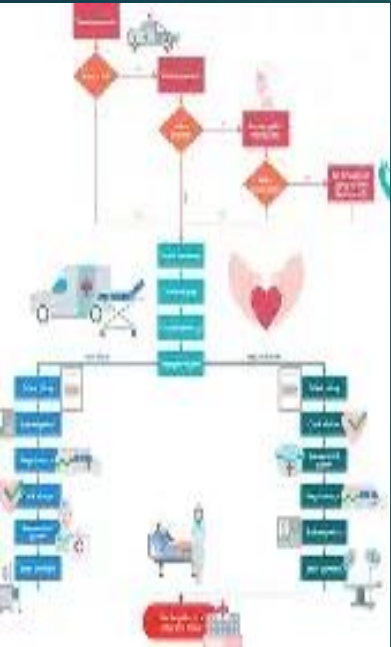
PROFESSOR CARAMANCION



INDUSTRY OVERVIEW

Healthcare Industry Today

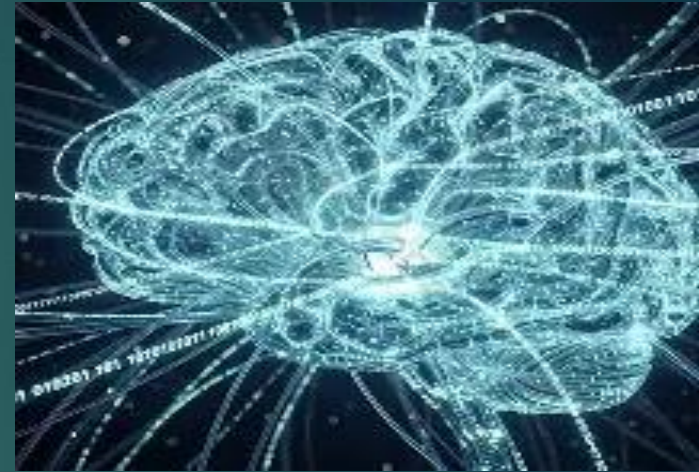
- ▶ Complex and data-driven
- ▶ High demand for services
- ▶ Resource and staffing challenges



WHAT IS AI?

Artificial Intelligence

- ▶ Learns patterns from data
- ▶ Makes predictions and decisions
 - ▶ Improves over time



KEY PROBLEMS IN HEALTHCARE

Major Challenges

- ▶ Delayed diagnosis
- ▶ Inefficient workflows
- ▶ Clinician burnout



AI SOLUTION 1: IMAGE

ANALYSIS

Image Analysis



- ▶ Analyzes medical images
 - ▶ Detects abnormalities
 - ▶ Supports early diagnosis



IMAGE ANALYSIS

EXAMPLE

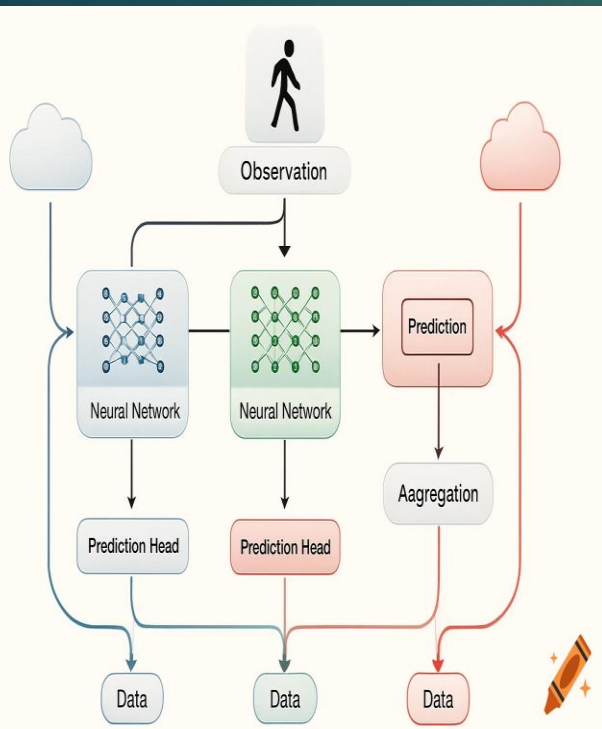
Example

- ▶ AI detects early-stage disease
 - ▶ Alerts clinicians
 - ▶ Speeds up diagnosis



AI SOLUTION 2: PREDICTIVE ANALYTICS

Predictive Analytics



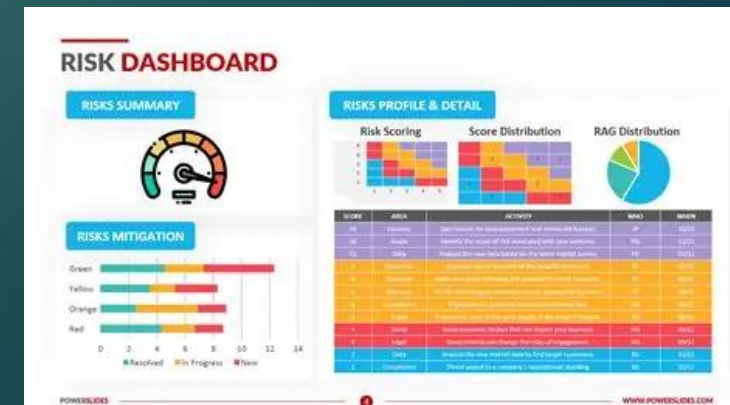
- ▶ Uses patient data
- ▶ Predicts disease risk
- ▶ Supports prevention

PREDICTIVE ANALYTICS

EXAMPLE

Example

- ▶ Identifies high-risk patients
- ▶ Enables early intervention
- ▶ Reduces hospitalizations



AI SOLUTION 3: PROCESS OPTIMIZATION

Process Optimization



- ▶ Improves workflows
- ▶ Reduces inefficiencies
- ▶ Saves time and resources

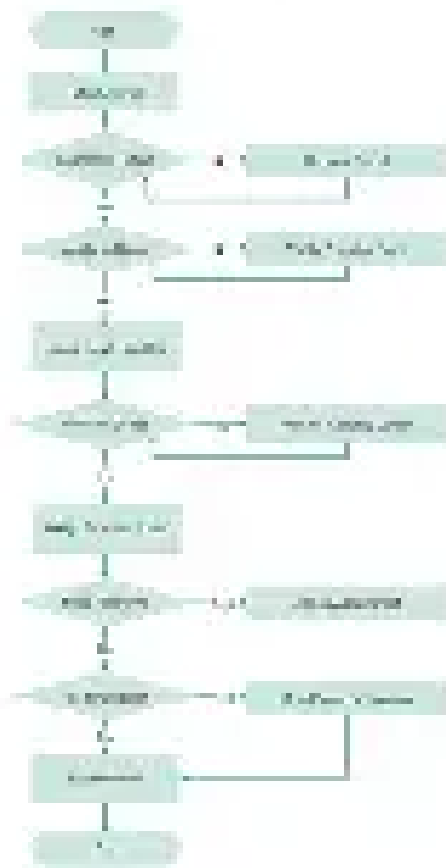
PROCESS OPTIMIZATION

EXAMPLE

Example

- ▶ Automated scheduling
- ▶ Reduced wait times
- ▶ Improved efficiency

Figure 1: Process Flowchart for Patient Scheduling



RESPONSIBLE AI

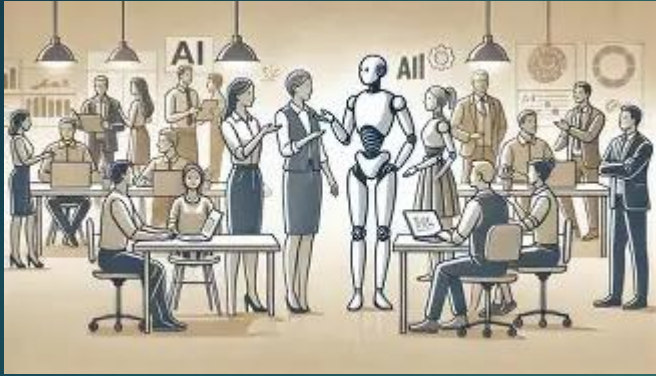
Ethical Concerns

- ▶ Wrong predictions
 - ▶ Bias
 - ▶ Privacy



RESPONSIBLE AI SOLUTIONS

Solutions



- ▶ Human oversight
 - ▶ Diverse data
 - ▶ Transparency





FUTURE IMPACT

Future of Healthcare AI

Benefits

- Faster diagnosis
- Better outcomes
- Increased efficiency

Challenges

- Trust
- Regulation
- Over-reliance

CONCLUSION

Key Takeaway

*AI will not replace healthcare—
It will transform it*

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